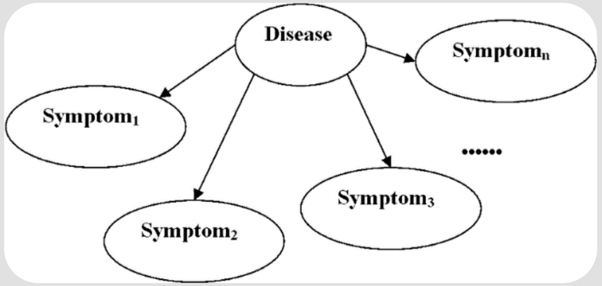
Probability and Statistics

Introduction

- Classification: Bayesian Methods
- Clustering
- Regression


Disease
Symptom$_n$
Symptom$_1$
......
Symptom$_3$
Symptom$_2$


90% accurate
80% accurate
50% accurate
Mon
Tue
Wed
Thu
Fri
Sat
Sun
Mon
Tue
Wed
?
?
?
76°
74°
70°
70°
71°
76°
75°


Credit Card Fraud Detection
3225

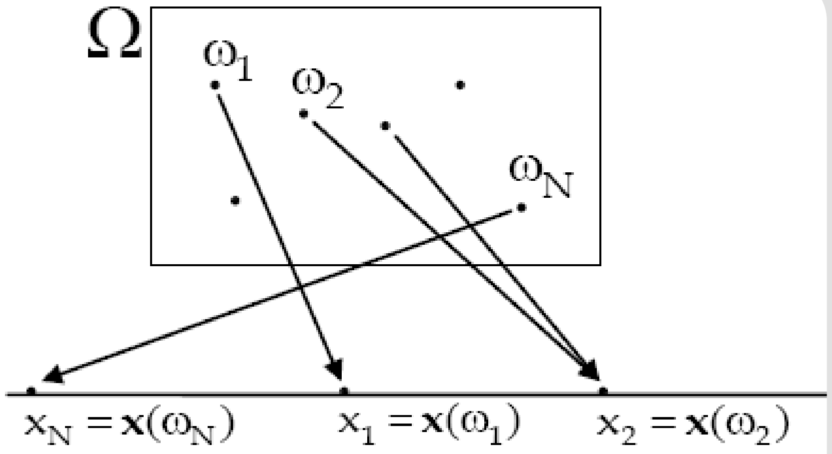
Probability

- Random experiment
- Sample space
- Event

Probability

- Conditional probability
- Total probability theorem
- Bayes theorem
- Independent events

Random Variables (RVs)



Discrete RVs

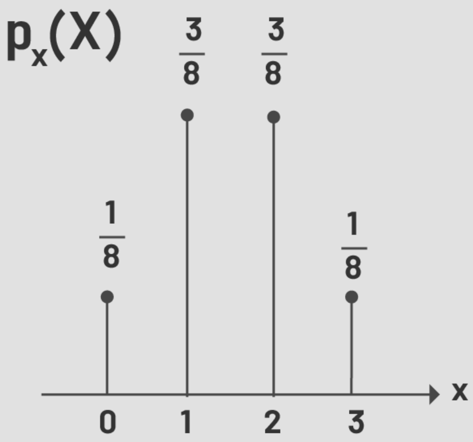
Continuous RVs

Probability Mass Function(PMF)

- $X : \{x_1, x_2, \ldots\}$

$P_1 \quad P_2$

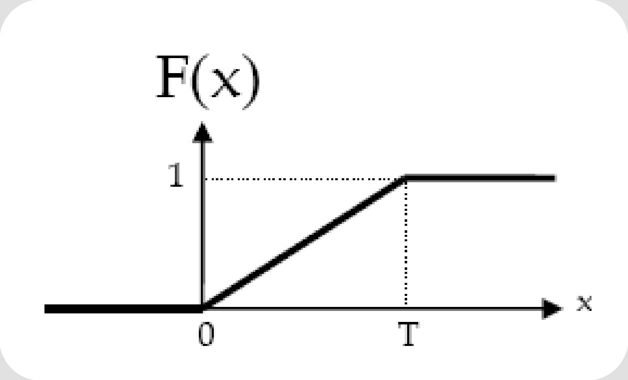
$$P_x(x) = \text{Prob}\{x = x\} = \begin{cases} p_i & x = x_i \\ 0 & \text{otherwise} \end{cases}$$



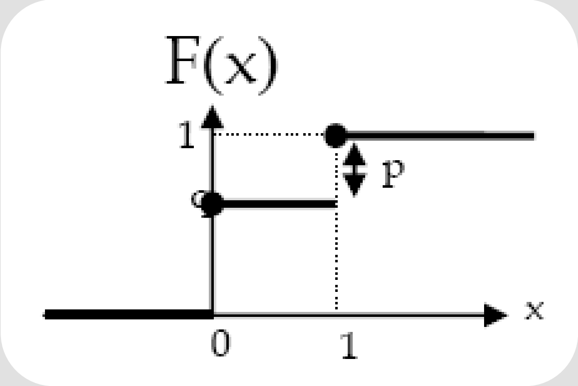
Cumulative Distribution Function(CDF)

$$\boxed{F_x(x) = \mathrm{Prob}\{x \le x\}}$$

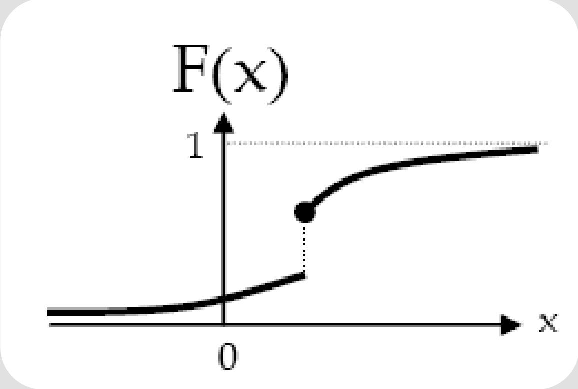
CDFs (continuous, discrete, mixed)



continuous



discrete

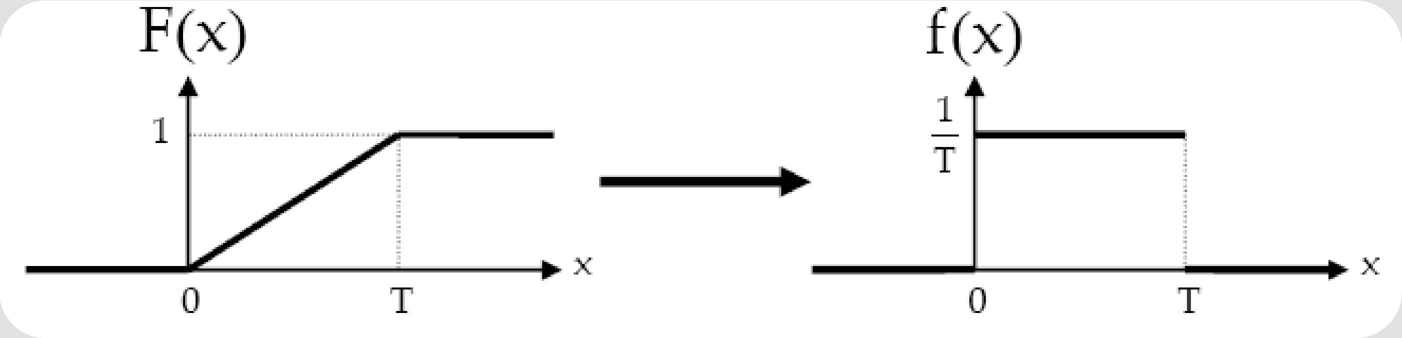


Mixed

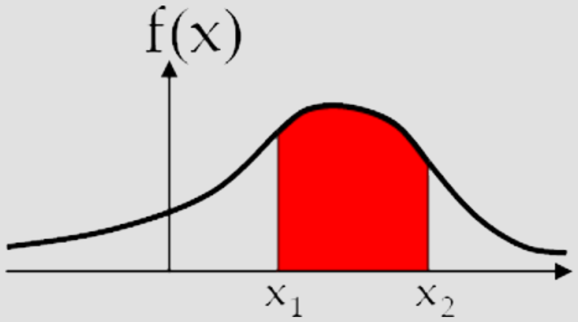
Probability Density Function(PDF)

$$\boxed{f_x(x) = \frac{dF_x(x)}{dx}}$$

$$F(x) = \int_{-\infty}^{x} f(u)du$$



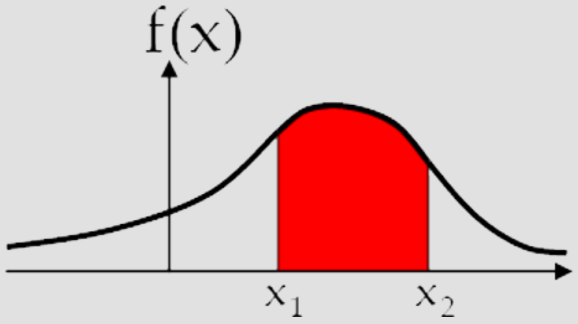
1) $P\{x_1 \leq x \leq x_2\} = \int_{x_1}^{x_2} f_x(x)dx$



$$1)\ P\{x_1 \leq x \leq x_2\} = \int_{x_1}^{x_2} f_x(x)dx$$

$$2)\quad f(x) \geq 0$$

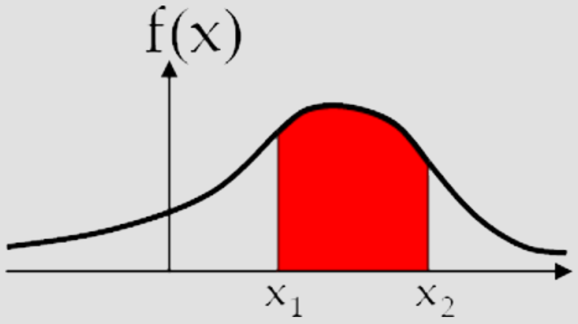
$$3)\ \int_{-\infty}^{+\infty} f(x)dx = 1$$



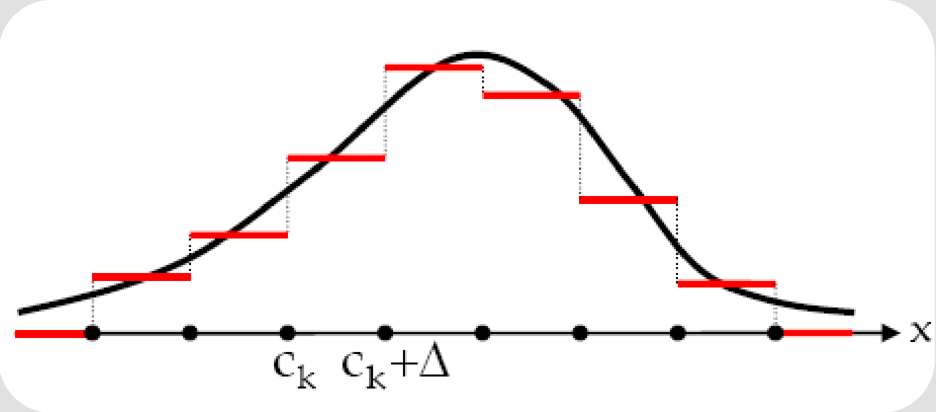
1) $P\{x_1 \leq x \leq x_2\} = \int_{x_1}^{x_2} f_x(x)dx$

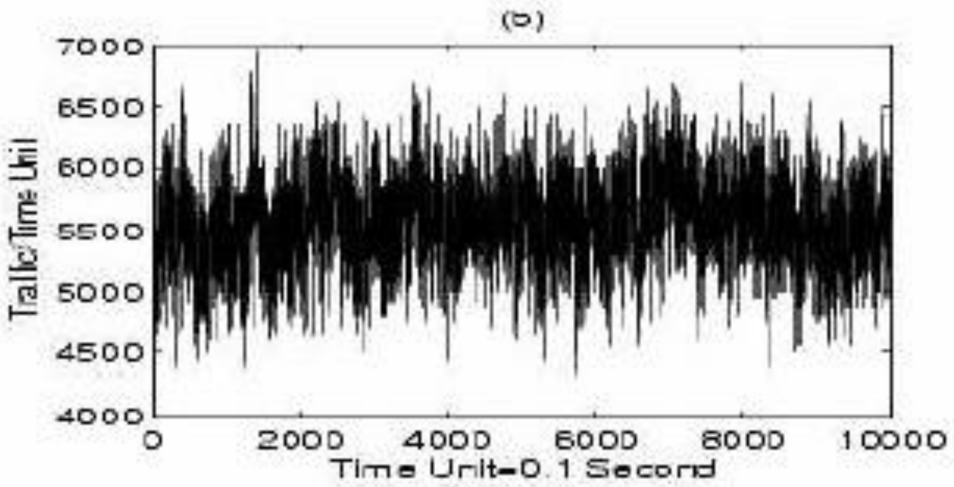
2) $f(x) \geq 0$

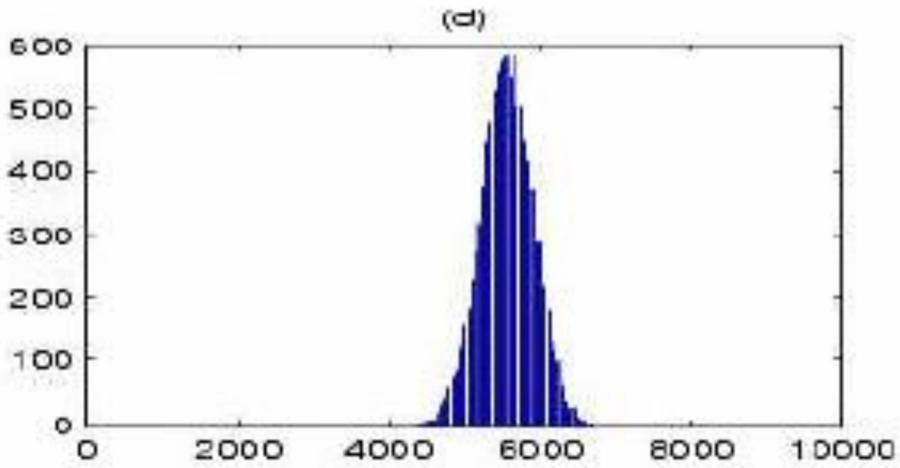
3) $\int_{-\infty}^{+\infty} f(x)dx = 1$



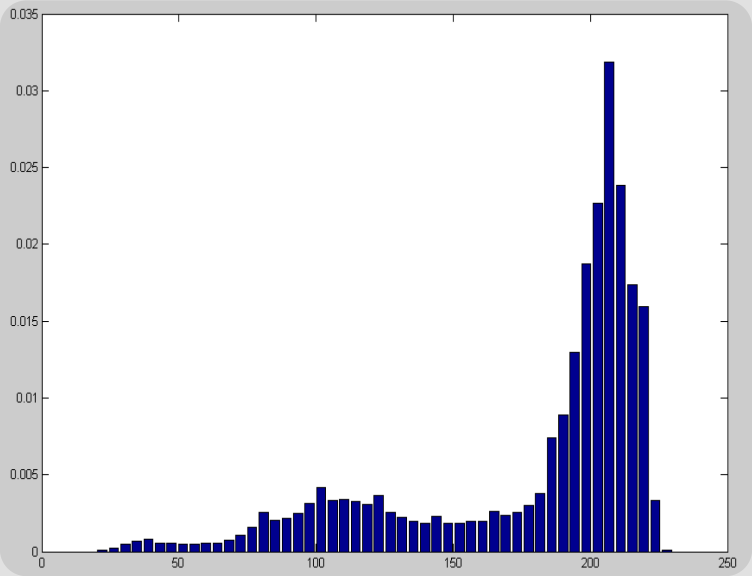
Experimental PDF(Histogram)




(b)
Traffic/Time Unit
7000
6500
6000
5500
5000
4500
4000
0
2000
4000
6000
8000
10000
Time Unit=0.1 Second

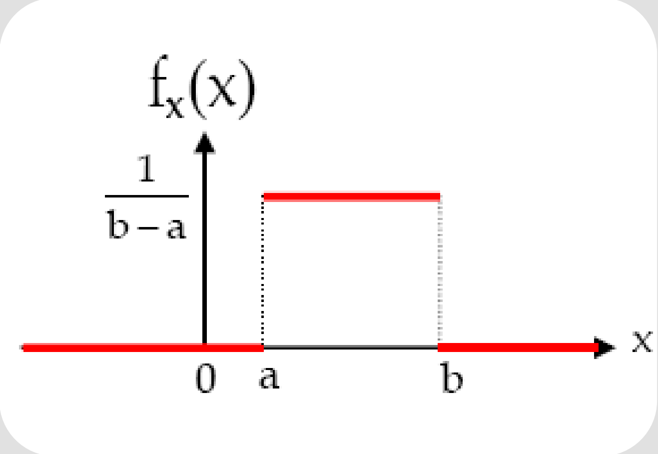

(d)
600
500
400
300
200
100
0
0
2000
4000
6000
8000
10000

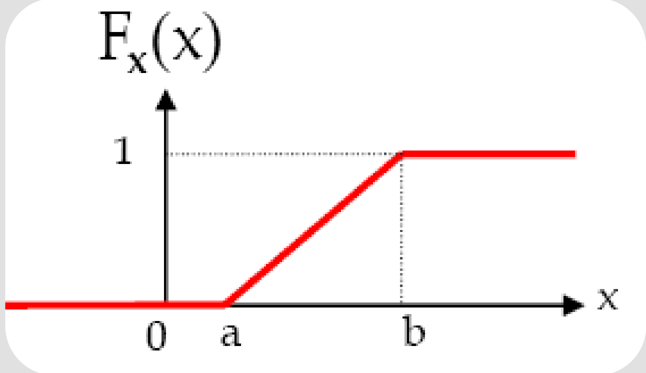

F-16
01568
U.S. AIR FORCE


0.035
0.03
0.025
0.02
0.015
0.01
0.005
0
0
50
100
150
200
250

Important Distributions

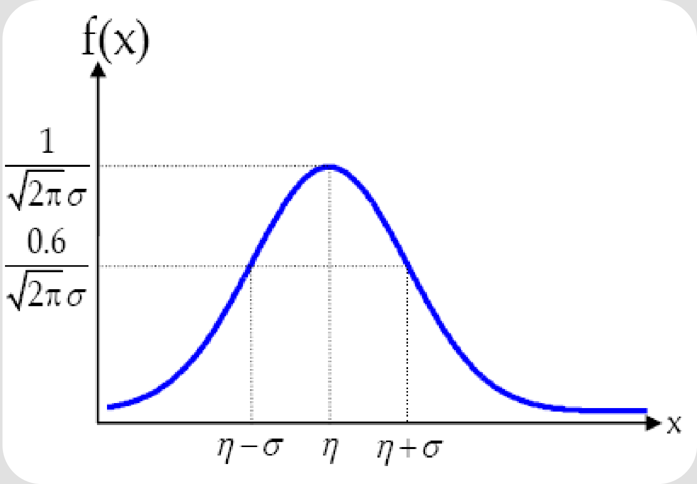
Uniform Distribution

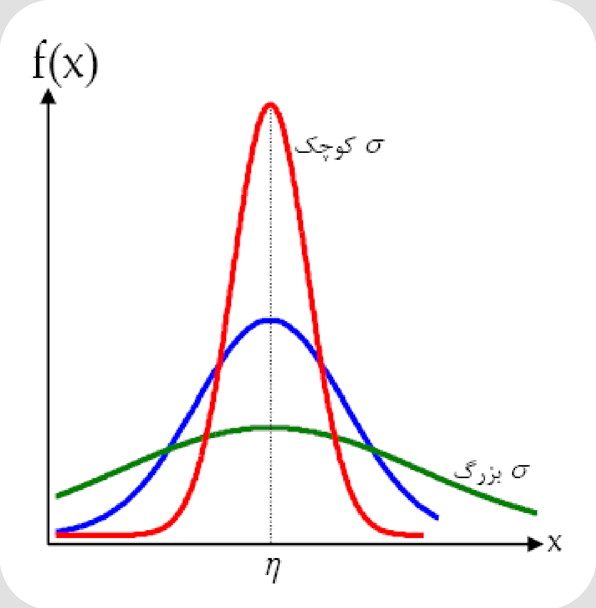


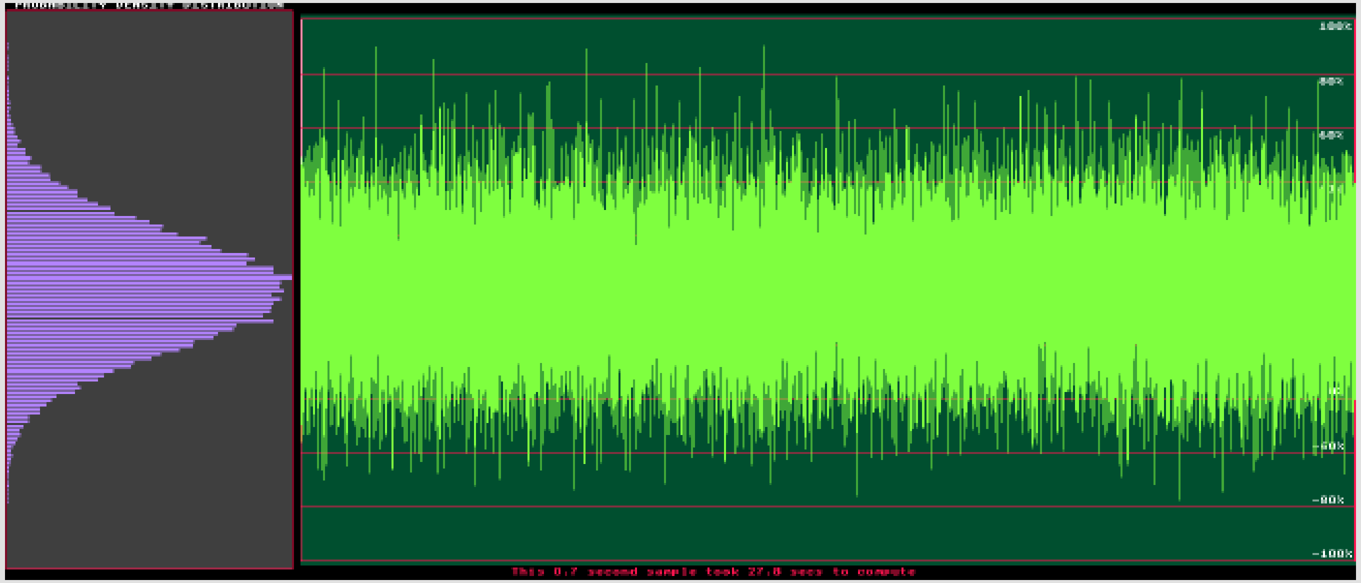


Gaussian Distribution(Normal)

$$f_x(x) = \frac{1}{\sqrt{2\pi\sigma}} e^{-\frac{(x-\eta)^2}{2\sigma^2}}$$





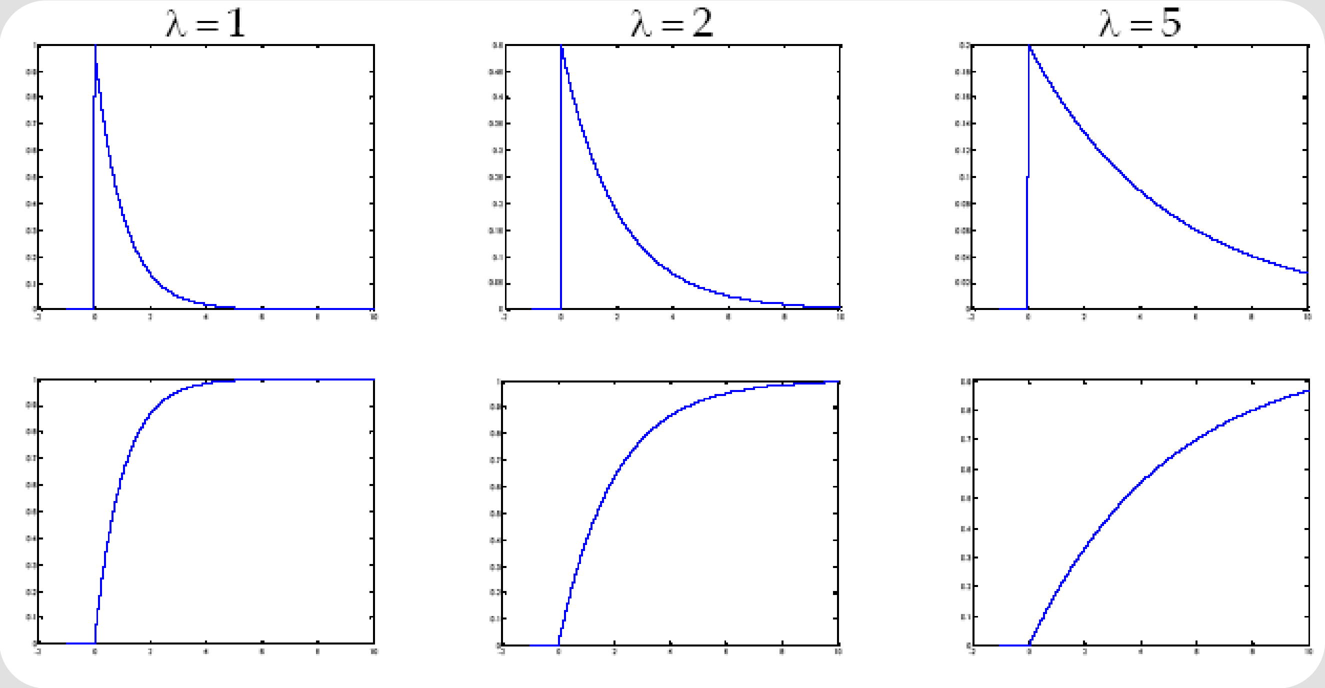


$$P\{\eta - \sigma \le \mathbf{x} \le \eta + \sigma\} \square 0.683$$
$$P\{\eta - 2\sigma \le \mathbf{x} \le \eta + 2\sigma\} \square 0.954$$
$$P\{\eta - 3\sigma \le \mathbf{x} \le \eta + 3\sigma\} \square 0.997$$

Exponential Distribution

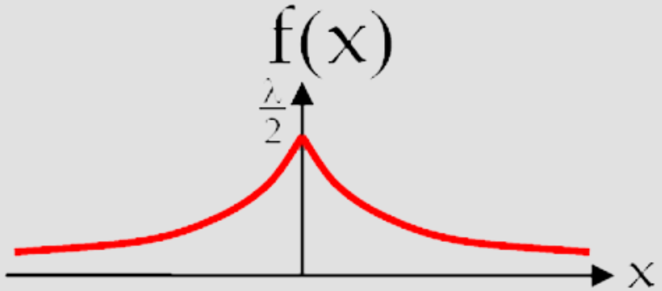
$$\boxed{f_x(x) = \lambda e^{-\lambda x} : x \geq 0}$$

$$\mathrm{F_x(x) = (1 - e^{-\lambda x})u(x)}$$



Laplace Distribution

$$f(x) = \frac{\lambda}{2} e^{-\lambda|x|}$$

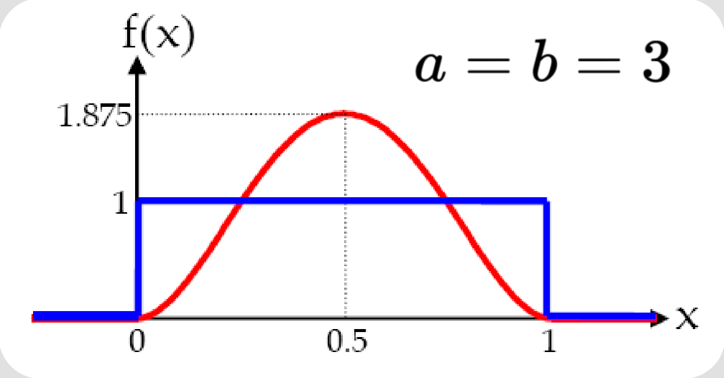


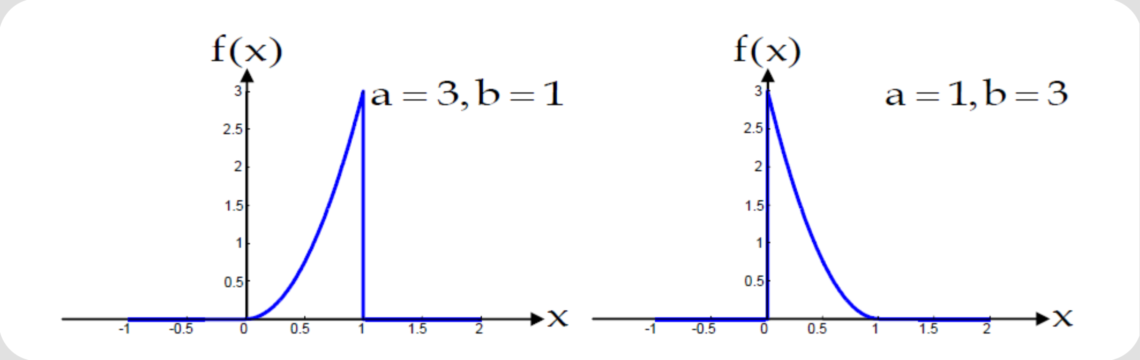
speech recognition to model priors on DFT coefficients

Beta Distribution

$$f(x) = \begin{cases} \frac{1}{\beta(a,b)} x^{a-1}(1-x)^{b-1} & 0 \le x \le 1 \\ 0 & \text{otherwise} \end{cases}$$

$$\beta(a,b) = \int_0^1 x^{a-1}(1-x)^{b-1}dx$$





Generalized Gaussian Distribution

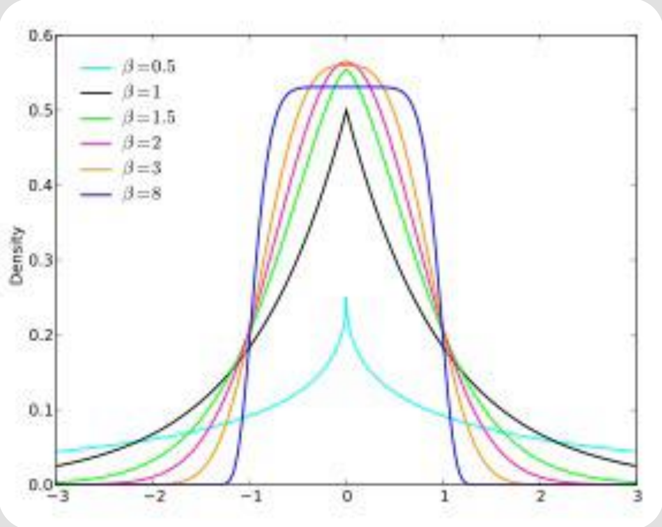
Location parameter

Shape parameter

$$f(x) = \frac{\beta}{2\alpha\Gamma(1/\beta)} e^{-(|x-\mu|/\alpha)^{\beta}}$$

Scale parameter

Probability density function



This family includes the normal distribution when $\beta = 2$ (with mean μ and variance $\frac{\alpha^2}{2}$) and it includes the Laplace distribution when $\beta = 1$

This family allows for tails that are either heavier than normal (when $\beta < 2$) or lighter than normal

Generalized Gaussian Distribution

- Applications: Discrete Cosine Transform (DCT) coefficients, the wavelet transform coefficients, pixels difference, video and geometry compression, watermarking, etc.

- GGD is also known in economy as Generalized Error Distribution (GED).

"A model PDF can characterize the statistical behavior of a signal. For multimedia signals, the generalized Gaussian distribution is often used."
Jens-Rainer "Multimedia Communication Technology".

Gaussian Mixture

- We cannot use one single Gaussian distribution to model real data sets.
- A linear combination of Gaussians can give rise to very complex densities densities.
- By using a sufficient number of Gaussians, and by adjusting their parameters as well as the coefficients in the linear combination, almost any continuous density can be approximated.

Gaussian Mixture

- Let's consider a random process x_i that follows a Gaussian mixture distribution with M component

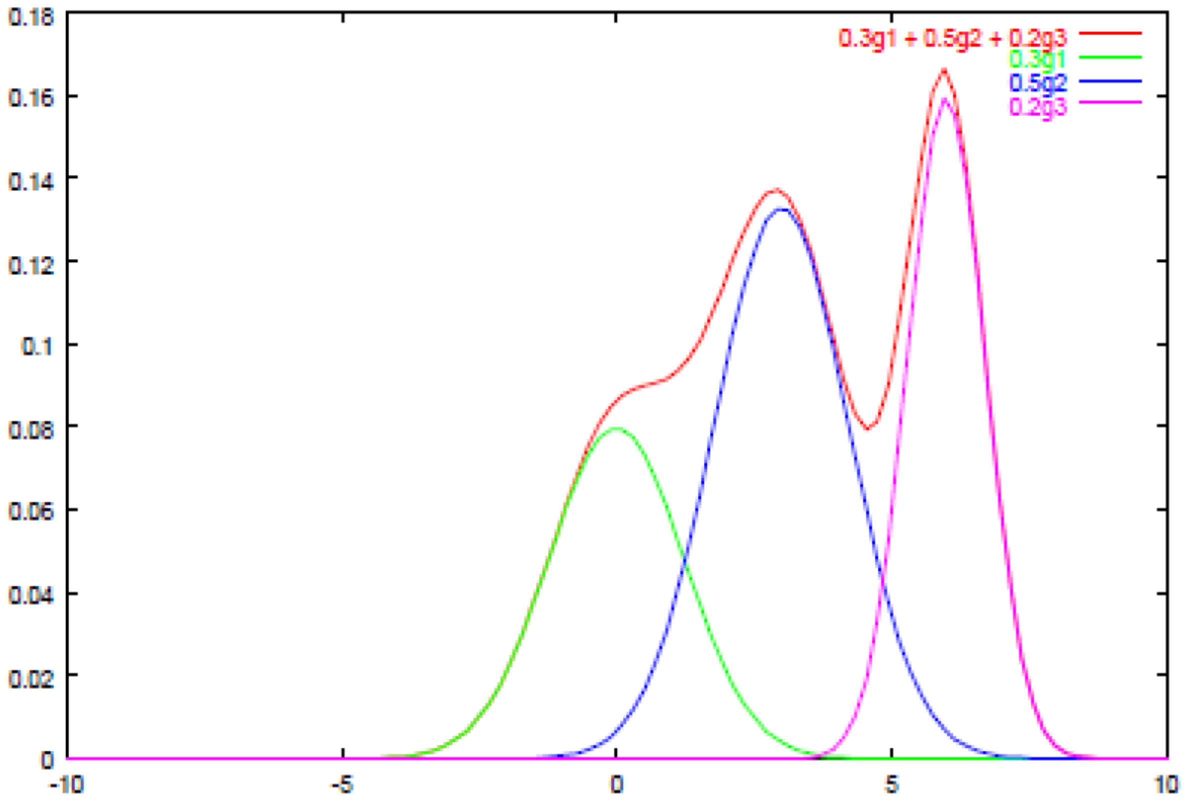
$$x_i \sim GM(w, \mu, \sigma)$$

$$\mu = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_M \end{pmatrix} \sigma = \begin{pmatrix} \sigma_1 \\ \sigma_2 \\ \vdots \\ \sigma_M \end{pmatrix}, w = \begin{pmatrix} w_1 \\ w_2 \\ \vdots \\ w_M \end{pmatrix}$$

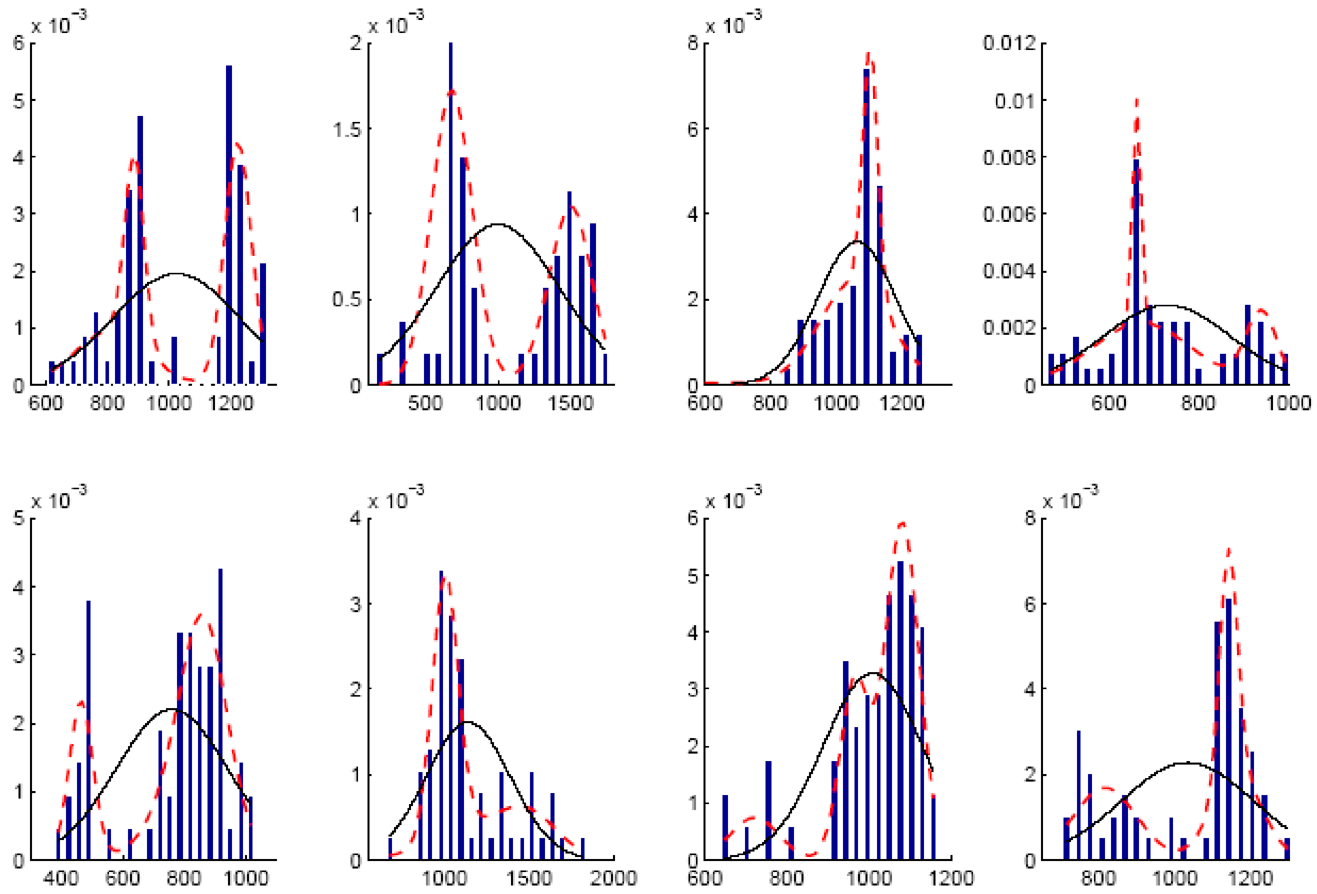
$$\boxed{f_{x_i}(x_i) = \sum_{j=1}^{M} \frac{w_j}{\sqrt{2\pi\sigma_j}} \exp \frac{-(x_i - m_j)^2}{2\sigma_j^2}}$$

Gaussian Mixture

Gaussian Mixture



Gaussian Mixture



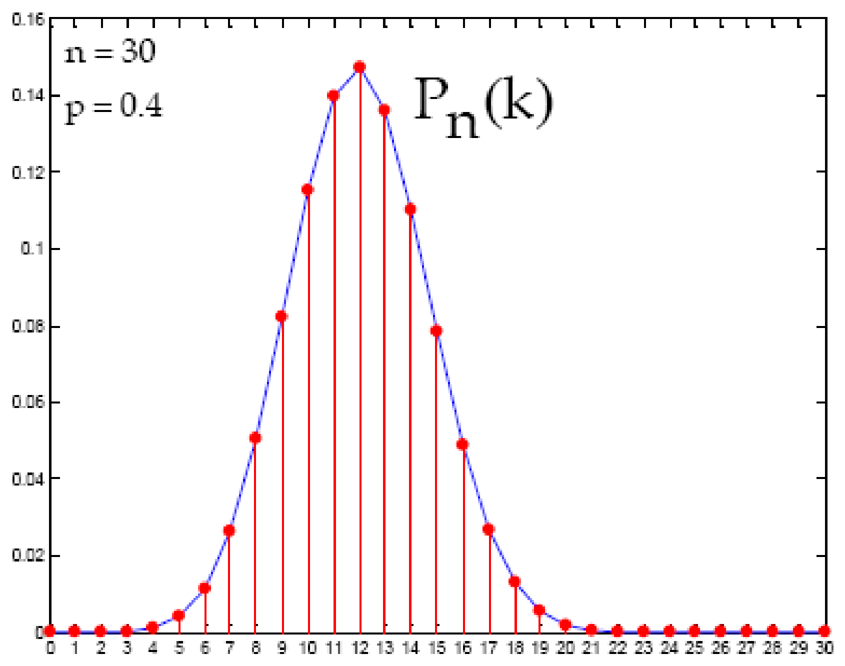
Discrete Distributions

Binomial Distribution

$$P(k) = P\{x = k\} = \binom{\mathbf{n}}{k} p^{\mathrm{k}} q^{\mathrm{n-k}} : \mathbf{k} = 0, 1, 2, \ldots, \mathbf{n}$$

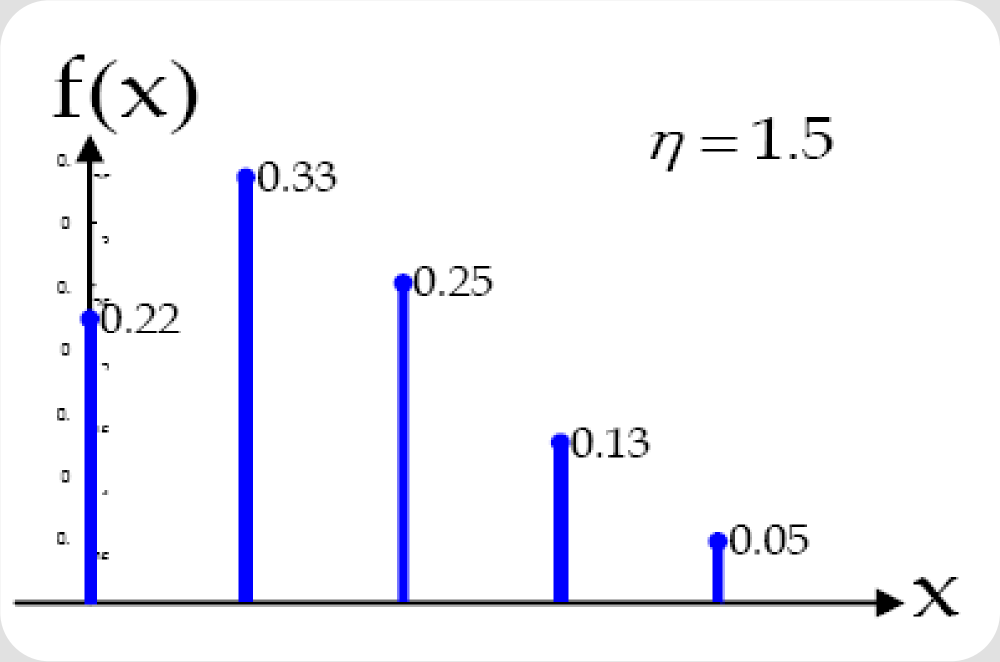
Binomial Distribution

$$P(k) = P\{x = k\} = \binom{\mathbf{n}}{k} p^{\mathrm{k}} q^{\mathrm{n-k}} : \mathbf{k} = 0, 1, 2, \ldots, \mathbf{n}$$



Poisson Distribution

$$P\{x = k\} = P(k) = e^{-\eta}\frac{\eta^k}{k!} \; : k = 0, 1, 2, \ldots$$



Mean and Variance

Mean(expectation)

$$\mathrm{E}(\mathbf{x}) = \sum_{\mathrm{i}} \mathrm{x_i} \mathrm{P_x}(\mathrm{x_i})$$

$$\eta = \mathrm{E}(\mathbf{x}) = \int_{-\infty}^{+\infty} x \mathrm{f}(\mathrm{x}) \mathrm{d}x$$

Symmetric Distribution $\forall x : f_x(a+x) = f_x(a-x) \implies E(x) = a$

$$y = g(x) \quad E(y) = \int_{-\infty}^{+\infty} y f_y(y) dy = \int_{-\infty}^{+\infty} g(x) f_x(x) dx$$

$$E(y) = \sum_i g(x_i) P_x(x_i)$$

Mean(expectation)

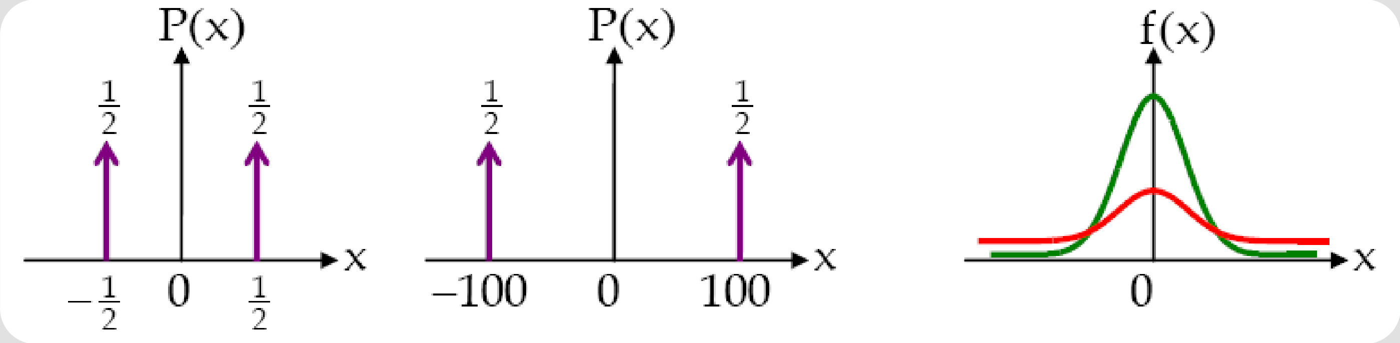
$$\mathrm{E}\left(\sum_{i=1}^{n} \mathrm{a_i g_i(x)}\right) := \sum_{\mathrm{i}=1}^{\mathrm{n}} \mathrm{a_i E(g_i(\mathbf{x}))}$$

$$P\{x \geq a\} = 1 \Longrightarrow E(x) \geq a$$

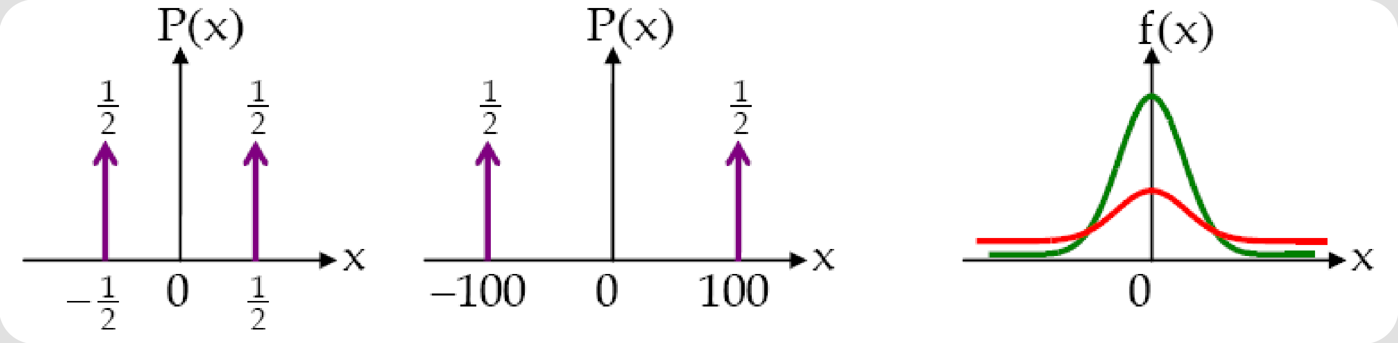
Mean(expectation)

$$\mathrm{E}\left(\sum_{i=1}^{n} \mathrm{a_i g_i}(\mathrm{x})\right) := \sum_{i=1}^{\mathrm{n}} \mathrm{a_i E}(\mathrm{g_i}(\mathbf{x}))$$

Variance



Variance



$$\mathrm{var}(\mathbf{x}) = \mathrm{E}\left([\mathbf{x} - \mathrm{E}(\mathbf{x})]^2\right)$$

$$\sigma^2 = \mathrm{E}\left((\mathbf{x} - \eta)^2\right)$$

Variance

$$\sigma^2 = \mathrm{E}\left(\mathbf{x}^2\right) - \mathrm{E}^2(\mathbf{x})$$

$$\mathrm{y} = \mathrm{ax} + \mathrm{b} \longrightarrow \sigma_{\mathrm{y}}^2 = \mathrm{a}^2 \sigma_{\mathrm{x}}^2$$

$$\operatorname{var}(\mathbf{x}) = 0 \longrightarrow \mathrm{P}\{\mathbf{x} = \eta\} = 1$$

1. Sample Mean

$$\bar{X} = \frac{1}{n}(X_1 + \cdots + X_n) = \frac{1}{n}\sum_{i=1}^{n} X_i$$

2. Sample Variance

$$S^2 = \frac{1}{n}\sum_{i=1}^{n}\left(X_i - \bar{X}\right)^2$$

Moments

Moments of a RV

$$m_n = E(\mathbf{x}^n) = \int_{-\infty}^{+\infty} x^n f(x)dx$$

$$\mu_n = E((\mathbf{x}-\eta)^n) = \int_{-\infty}^{+\infty} (x-\eta)^n f(x)dx$$

$$\eta = m_1, \sigma^2 = \mu_2, \mu_1 = 0, \mu_0 = m_0 = 1$$

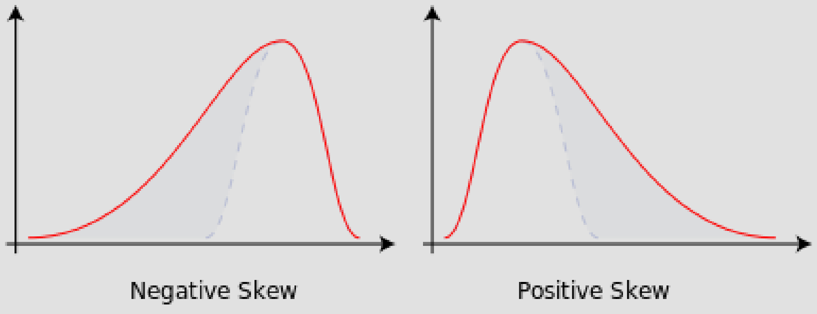
Sample Central Moments

$$M_r = \frac{1}{n}\sum_{i=1}^{n}(X_i - \bar{X})^r$$

Skewness

$$S = \frac{\mu_3}{\sigma^3} = \frac{\mathrm{E}\left[(X-\mu)^3\right]}{\left(\mathrm{E}\left[(X-\mu)^2\right]\right)^{3/2}}$$

Skewness is a measure of symmetry, or more precisely, the lack of symmetry.



Sample skewness

For a sample of n values the sample skewness is

$$\frac{\frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})^3}{\left(\frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})^2\right)^{3/2}}$$

kurtosis

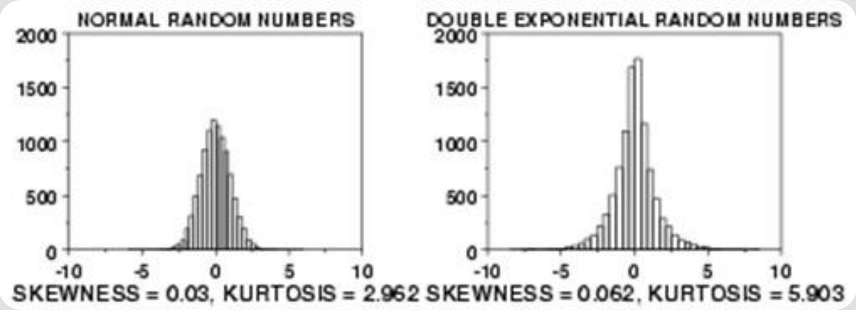
$$K = \frac{\mathrm{E}\left[(X-\mu)^4\right]}{\left(\mathrm{E}\left[(X-\mu)^2\right]\right)^2} = \frac{\mu_4}{\sigma^4}$$

data sets with high kurtosis tend to have a distinct peak near the mean, decline rather rapidly, and have heavy tails.

For a sample of n values the sample kurtosis is

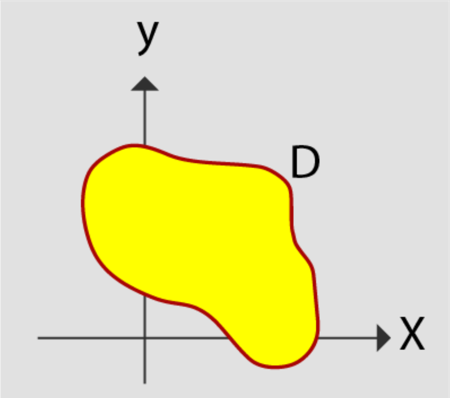
$$\frac{\frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})^4}{\left(\frac{1}{n}\sum_{i=1}^{n}(x_i-\bar{x})^2\right)^2}$$

Sample excess kurtosis: Kurtosis-3



Two Random Variables

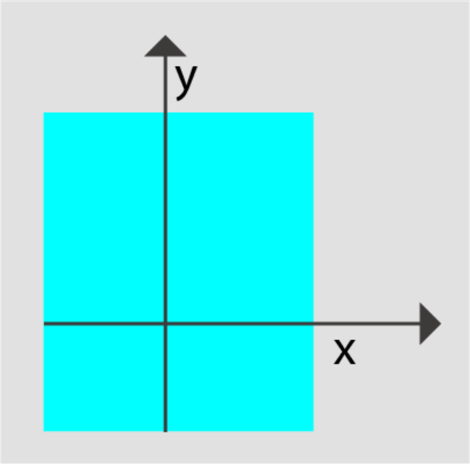
Two random variables



$$\{\mathbf{x} = x_1, \mathbf{y} = y_1\} \quad \{\mathbf{x} \le x_0, \mathbf{y} \le y_0\}$$

$$\{(\mathbf{x}, \mathbf{y})\} \in D$$

Joint CDF



$$F_{xy}(x, y) = P\{x \leq x, y \leq y\}$$

characteristics

$$F_{xy}(x, y) = \sum_{x_i \leq x} \sum_{y_j \leq y} P_{xy}(x_i, y_j)$$

$$2) \left\{ \begin{array}{l} F_{xy}(x, +\infty) = P\{x \leq x\} = F_x(x) \\ F_{xy}(+\infty, y) = P\{y \leq y\} = F_y(y) \end{array} \right\} : (\text{Marginal CDF})$$

Joint pdf

definition

$$f_{xy}(x,y) = \frac{\partial^2 F_{xy}(x,y)}{\partial x \partial y}$$

$$F_{xy}(x,y) = \int_{-\infty}^{x} \int_{-\infty}^{y} f(u,v)dudv = P\{x \le x, y \le y\}$$

characteristics

$$P\{(x,y) \in D\} = \iint_D f_{xy}(x,y)dxdy$$

$$f_{xy}(x,y)dxdy = P\{x < x \le x + dx, y < y \le y + dy\}$$

Joint pdf

$$\int_{-\infty}^{+\infty}\int_{-\infty}^{+\infty} f(x,y)dxdy = F(+\infty,+\infty) = 1$$

Marginal pdf

$$f_x(x) = \int_{-\infty}^{+\infty} f_{xy}(x,y)dy : x \qquad f_y(y) = \int_{-\infty}^{+\infty} f_{xy}(x,y)dx : y$$

Independency

$$\begin{cases} P\{x \le x, y \le y\} = P\{x \le x\}P\{y \le y\} \\ \mathrm{F_{xy}(x, y) = F_x(x)F_y(y)} \\ \mathrm{f_{xy}}(x, y) = \mathrm{f_x}(x)\mathrm{f_y}(\mathrm{y}) \end{cases}$$

Functions of two independent variables

Discrete Case

$$\underbrace{P\{\mathbf{x} = x_i, y = y_j\}}_{P_{xy}(x_i, y_j)} = \underbrace{P\{\mathbf{x} = x_i\}}_{P_x(x_i)}\underbrace{P\{y = y_j\}}_{P_y(y_j)}$$

Function of two RVs

$$z = g(x, y)$$

$$\mathrm{E}(\mathrm{g}(\mathrm{x}, \mathrm{y})) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} g(x, y) f_{xy}(x, y) dx dy$$

$$E\left[\sum_{\mathrm{i}=1}^{\mathrm{n}} \mathrm{a_i g_i(x, y)}\right] = \sum_{\mathrm{i}=1}^{\mathrm{n}} \mathrm{a_i E(g_i(x, y))}$$

Function of two RVs

Mean of two RVs

Variance of two RVs

New parameter

$$\operatorname{cov}(\mathbf{x}, \mathbf{y}) = \mu_{\mathrm{xy}} = \mathrm{E}[(\mathbf{x} - \eta_{\mathrm{x}})(\mathbf{y} - \eta_{\mathrm{y}})]$$

Function of two RVs

characteristics

$$\operatorname{cov}(x, y) = E(xy) - E(x)E(y)$$

$$\operatorname{cov}(\mathbf{x}, \mathbf{x}) = \operatorname{var}(\mathbf{x})$$

$$\operatorname{var}(\mathbf{x} + \mathbf{y}) = \operatorname{var}(\mathbf{x}) + \operatorname{var}(\mathbf{y}) + 2\operatorname{cov}(\mathbf{x}, \mathbf{y})$$

$$z = ax + by + c$$

$$\operatorname{var}(\mathbf{z}) = \mathrm{a}^2 \operatorname{var}(\mathbf{x}) + \mathrm{b}^2 \operatorname{var}(\mathbf{y}) + 2\mathrm{abcov}(\mathbf{x}, \mathbf{y})$$

Correlation Coefficient

$$r_{xy} = \frac{\operatorname{cov}(x, y)}{\sqrt{\operatorname{var}(x)\operatorname{var}(y)}} = \frac{\mu_{xy}}{\sigma_x \sigma_y}$$

$$-1 \leq r_{xy} \leq 1$$

Correlation Coefficient

Uncorrelated RVs

$$\sigma^2_{\mathrm{x+y}} = \sigma^2_{\mathrm{x}} + \sigma^2_{\mathrm{y}}$$

$$r_{xy} = 0 \Rightarrow \mathrm{cov}(\mathbf{x}, \mathbf{y}) = 0 \Rightarrow \mathrm{E(xy) = E(x)E(y)}$$

Orthogonal RVs

$$R_{xy} = E(xy) = 0$$

Independent RVs

Joint moments

$$m_{kr} = E(x^k y^r) = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} x^k y^r f_{xy}(x, y) dx dy$$

$$\mu_{xy} = E\left[(x - \eta_x)^k (y - \eta_y)^r\right] = \int_{-\infty}^{+\infty} \int_{-\infty}^{+\infty} (x - \eta_x)^k (y - \eta_y)^r f_{xy}(x, y) dx dy$$

$$\boxed{m_{11} = R_{xy}, \mu_{11} = \mathrm{cov}(\mathbf{x}, y) = \mu_{xy}, \mu_{20} = \sigma_x^2}$$

Jointly Normal

$$\mathbf{z} = \mathrm{ax} + \mathrm{b}y$$

$$\mathrm{f_{xy}(x, y)} = \frac{1}{2\pi\sigma_\mathrm{x}\sigma_\mathrm{y}\sqrt{1-\mathrm{r}^2}}\exp\left\{-\frac{1}{2(1-\mathrm{r}^2)}\left[\frac{(\mathrm{x}-\eta_\mathrm{x})^2}{\sigma_\mathrm{x}^2} - 2\mathrm{r}\frac{(\mathrm{x}-\eta_\mathrm{x})(\mathrm{y}-\eta_\mathrm{y})}{\sigma_\mathrm{x}\sigma_\mathrm{y}} + \frac{(\mathrm{y}-\eta_\mathrm{y})^2}{\sigma_\mathrm{y}^2}\right]\right\}$$

$$\eta_\mathrm{y} \cdot \eta_\mathrm{x} \cdot \sigma_\mathrm{y} \cdot \sigma_\mathrm{x}$$

uncorrelated

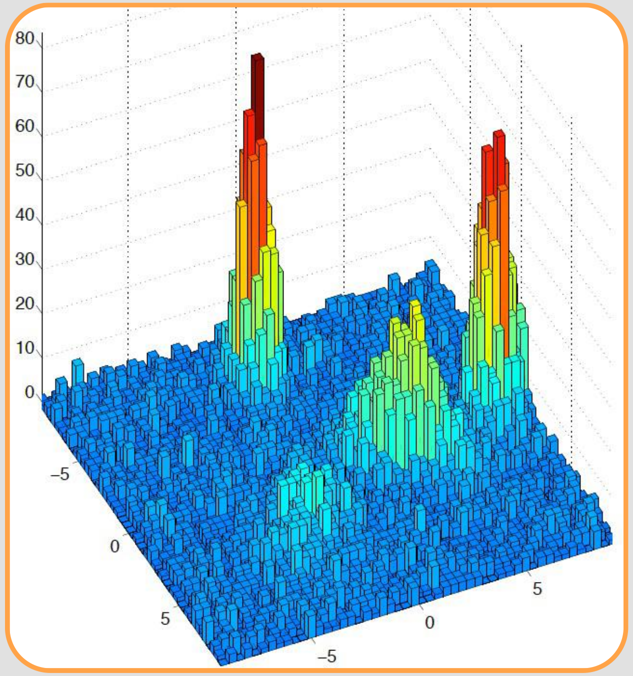
↑

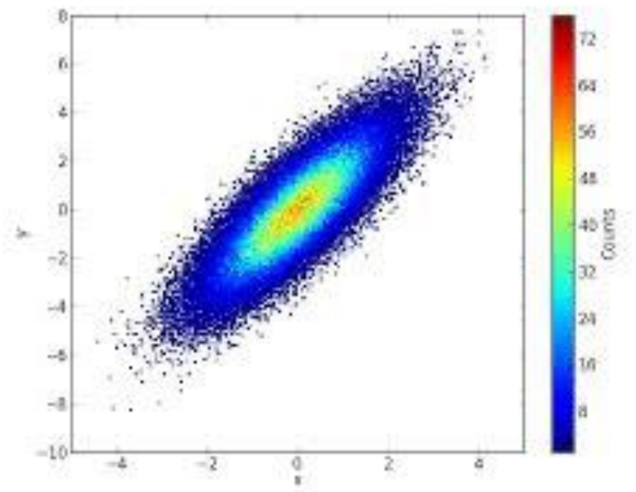
Independency

Jointly Normal

Working with samples

2D histogram





2D histogram

Sample mean

Sample Variance

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

Sample Correlation Coefficients

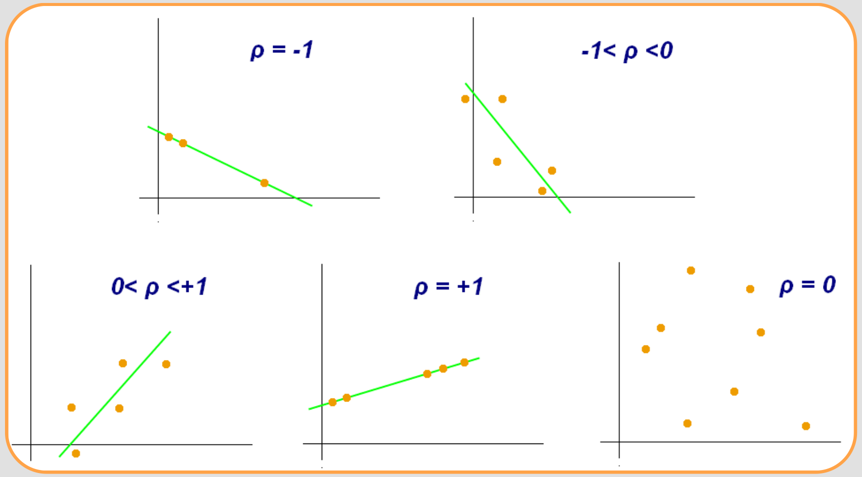
2D histogram

Sample mean

Sample Variance

$$r = \frac{\sum_{i=1}^{n}(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^{n}(x_i - \bar{x})^2}\sqrt{\sum_{i=1}^{n}(y_i - \bar{y})^2}}$$

Sample Correlation Coefficients



Random Vector

Maryam Amirmazlaghani

Amirkabir University of Technology
Department of Computer Engineering and Information Theory

Joint PMF

$$\underline{\mathbf{x}} = \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x_n} \end{bmatrix} \qquad P_{\underline{x}}(\underline{x}) = P_{x_1 x_2 \cdots x_n}(x_1, x_2, \ldots, x_n) = P\{\mathbf{x}_1 = x_1, \mathbf{x}_2 = x_2, \ldots, \mathbf{x}_n = x_n\}$$

Joint CDF

$$\mathrm{F}_{\underline{x}}(\underline{x}) = \mathrm{F}_{\mathrm{x_1 x_2 \cdots x_n}}(\mathrm{x_1, x_2, \ldots, x_n}) = \mathrm{P}\{\mathbf{x}_1 \leq \mathrm{x}_1, \mathbf{x}_2 \leq \mathrm{x}_2, \ldots, \mathbf{x}_\mathrm{n} \leq \mathrm{x_n}\}$$

Joint FDF

$$\mathrm{f}_{\underline{x}}(\underline{x}) = f_{x_1 x_2 \cdots x_n}(x_1, x_2, \ldots, x_n)$$

$$= \frac{\partial^{\mathrm{n}}}{\partial \mathrm{x}_1 \partial \mathrm{x}_2 \cdots \partial \mathrm{x}_{\mathrm{n}}} \mathrm{F}_{\underline{x}}(\underline{x}) = \frac{\partial^{\mathrm{n}}}{\partial \mathrm{x}_1 \partial \mathrm{x}_2 \cdots \partial \mathrm{x}_{\mathrm{n}}} \mathrm{F}_{\mathrm{x}_1 \mathrm{x}_2 \cdots \mathrm{x}_{\mathrm{n}}}(\mathrm{x}_1, \mathrm{x}_2, \ldots, \mathrm{x}_{\mathrm{n}})$$

$$f_{x_2 x_3}(x_2, x_3) = \int_{-\infty}^{+\infty} f_{x_1 x_2 x_3}(x_1, x_2, x_3) dx_1$$

$$\mathrm{F}_{\mathrm{x}_1 \mathrm{x}_2 \cdots \mathrm{x}_{\mathrm{n}}}(\mathrm{x}_1, \mathrm{x}_2, \ldots, \mathrm{x}_{\mathrm{n}}) = \int_{-\infty}^{x_1} \int_{-\infty}^{x_2} \cdots \int_{-\infty}^{x_{\mathrm{n}}} \mathrm{f}_{\mathrm{x}_1 \mathrm{x}_2 \cdots \mathrm{x}_{\mathrm{n}}}(\mathrm{u}_1, \mathrm{u}_2, \ldots, \mathrm{u}_{\mathrm{n}}) \mathrm{d}\mathrm{u}\mathrm{u}_{\mathrm{n}} \cdots \mathrm{d}\mathrm{u}_2 \mathrm{d}\mathrm{u}_1$$

$$P\{(x_1, x_2, \ldots, x_n) \in D\} = \int \cdots \int f_D f_{x_1 x_2 \cdots x_n}(x_1, x_2, \ldots, x_n) dx_1 \cdots dx_n = \int_D f_{\underline{x}}(\underline{x}) d\underline{x}$$

Joint FDF

Independency

$$\begin{cases} F_{x_1x_2\cdots x_n}(x_1,x_2,\ldots,x_n)=F_{x_1}(x_1)F_{x_2}(x_2)\cdots F_{x_n}(x_n)=\prod_{i=1}^{n}F_{x_i}(x_i) \\ f_{x_1x_2\cdots x_n}(x_1,x_2,\ldots,x_n)=f_{x_1}(x_1)f_{x_2}(x_2)\cdots f_{x_n}(x_n)=\prod_{i=1}^{n}f_{x_i}(x_i) \end{cases}$$

$$\overbrace{\mathbf{x}_1,\mathbf{x}_2,\ldots,\mathbf{x}_m},\overbrace{\mathbf{x}_{m+1},\ldots,\mathbf{x}_n}$$

$$f_{x_1x_2\cdots x_n}(x_1,x_2,\ldots,x_n)=f_{x_1x_2\cdots x_m}(x_1,x_2,\ldots,x_m)f_{x_{m+1}\cdots x_n}(x_{m+1},\ldots,x_n)$$

Joint FDF

$$\underline{\mathbf{x}} = \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_n \end{bmatrix} \Rightarrow \eta_{\underline{\mathbf{x}}} = \mathrm{E}(\underline{\mathbf{x}}) = \begin{bmatrix} \mathrm{E}(\mathbf{x}_1) \\ \mathrm{E}(\mathbf{x}_2) \\ \vdots \\ \mathrm{E}(\mathbf{x}_\mathbf{n}) \end{bmatrix} = \begin{bmatrix} \eta_1 \\ \eta_2 \\ \vdots \\ \eta_\mathrm{n} \end{bmatrix}$$

$$\mu_{x_i x_j} = \mathrm{cov}(\mathbf{x}_i, x_j) = E[(\mathbf{x}_i - \eta_i)(\mathbf{x}_j - \eta_j)] : i, j : 1, 2, \ldots, n$$

$$\sigma_{\mathrm{ii}} = \mathrm{cov}(\mathbf{x}_\mathrm{i}, \mathbf{x}_\mathrm{i}) = \mathrm{var}(\mathbf{x}_\mathrm{i}) = \sigma_\mathrm{i}^2$$

Covariance Matrix

$$C_{\underline{\mathbf{x}}} = \mathrm{E}\left[(\underline{\mathbf{x}} - \eta_{\underline{\mathbf{x}}}) \cdot (\underline{\mathbf{x}} - \eta_{\underline{\mathbf{x}}})^\mathrm{T}\right]$$

Joint FDF

$$C_{\underline{\mathbf{x}}} = \mathrm{E}\left[(\underline{\mathbf{x}} - \eta_{\underline{\mathbf{x}}}) \cdot (\underline{\mathbf{x}} - \eta_{\underline{\mathbf{x}}})^{\mathrm{T}}\right] = E\left(\begin{bmatrix} \mathbf{x}_1 - \eta_1 \\ \mathbf{x}_2 - \eta_2 \\ \vdots \\ \mathbf{x}_{\mathrm{n}} - \eta_{\mathrm{n}} \end{bmatrix} \cdot \begin{bmatrix} \mathbf{x}_1 - \eta_1 & \mathbf{x}_2 - \eta_2 & \cdots & \mathbf{x}_{\mathbf{n}} - \eta_{\mathrm{n}} \end{bmatrix}\right)$$

$$= \begin{bmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1\mathrm{n}} \\ \sigma_{21} & \sigma_{22} & \cdots & \sigma_{2\mathrm{n}} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{\mathrm{n}1} & \sigma_{\mathrm{n}2} & \cdots & \sigma_{\mathrm{nn}} \end{bmatrix}$$

Joint FDF

Characteristics of Covariance Matrix

$$C = C^{\mathrm{T}} \quad \sigma_{\mathrm{ij}} = \sigma_{\mathrm{ji}}$$

Non-negative Definite $\quad \underline{a}^{\mathrm{T}} \mathcal{C}_{\underline{a}} \geq 0 \quad \det(C) \geq 0$

Joint FDF

Correlation Matrix

$$R_{\underline{\mathbf{x}}} = \mathrm{E}\left(\underline{\mathbf{x}} \cdot \underline{\mathbf{x}}^{\mathrm{T}}\right) \qquad C_{\underline{\underline{x}}} = R_{\underline{\underline{x}}} - \eta_{\underline{\underline{x}}} \cdot \eta_{\underline{\underline{x}}}^{\mathrm{T}}$$

Cross Covariance Matrix

$$\mathrm{C}_{\underline{\mathrm{x}}\underline{\mathrm{y}}} = \mathrm{cov}(\underline{\mathbf{x}}, \underline{\mathbf{y}}) = \mathrm{E}\left[(\underline{\mathbf{x}} - \eta_{\underline{\mathbf{x}}})\left(\underline{\mathbf{y}} - \eta_{\underline{\mathrm{y}}}\right)^{\mathrm{T}}\right]$$

Cross Correlation Matrix

$$\mathrm{R}_{\underline{x}\underline{\mathrm{y}}} = \mathrm{E}\left(\underline{\mathbf{x}} \cdot \underline{\mathrm{y}}^{\mathrm{T}}\right)$$

Joint FDF

Jointly Normal

$$\mathbf{z} = \sum_{i=1}^{n} a_i x_i$$

$$f_{\underline{x}}(\underline{x}) = \frac{1}{(2\pi)^{\frac{1}{2}}\sqrt{\det(C_{\underline{\mathbf{x}}})}} \exp\left[-\frac{1}{2}(\underline{x} - \eta_{\underline{\mathbf{x}}})^{\mathrm{T}} \mathrm{C}_{\underline{\mathbf{x}}}^{-1} (\underline{\mathbf{x}} - \eta_{\underline{\mathbf{x}}})\right]$$

Uncorrelated

$$\sigma_{\mathrm{ij}} = 0 : \mathrm{i} \neq \mathrm{j} \quad \Rightarrow \mathrm{C}_{\underline{\mathrm{x}}} = \begin{bmatrix} \sigma_1^2 & 0 & \cdots & 0 \\ 0 & \sigma_2^2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & 0 \\ 0 & \cdots & 0 & \sigma_{\mathrm{n}}^2 \end{bmatrix}$$

$$\Rightarrow \mathrm{f}_{\underline{\mathrm{x}}}(\underline{\mathrm{x}}) = \frac{1}{(2\pi)^{\frac{\mathrm{n}}{2}}\sqrt{\prod_{\mathrm{i}=1}^{\mathrm{n}} \sigma_{\mathrm{i}}^2}} \exp\left\{-\frac{1}{2}\sum_{\mathrm{i}=1}^{\mathrm{n}} \frac{(\mathrm{x_i} - \eta_{\mathrm{i}})^2}{\sigma_{\mathrm{i}}^2}\right\} = \prod_{\mathrm{i}=1}^{\mathrm{n}} \mathrm{f}_{\mathrm{x_i}}(\mathrm{x_i})$$

Joint FDF

Independent Identically Distributed

$$\underline{\mathbf{x}} = \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_n \end{bmatrix} \qquad \mathrm{f}_{\underline{\mathbf{x}}}(\underline{\mathrm{x}}) = \mathrm{f}_{\mathrm{x}_1}(\mathrm{x}_1) \cdots \mathrm{f}_{\mathbf{x}_\mathrm{n}}(\mathrm{x}_\mathbf{n}) = \mathrm{f}_{\mathbf{x}}(\mathrm{x}_1) \cdots \mathrm{f}_{\mathbf{x}}(\mathrm{x}_\mathbf{n})$$

Central Limit Theorem (CLT)

$$\underline{\mathbf{x}} = \begin{bmatrix} \mathbf{x}_1 \\ \mathbf{x}_2 \\ \vdots \\ \mathbf{x}_\mathbf{n} \end{bmatrix} \qquad \mathbf{y} = \sum_{i=1}^{n} x_i$$

$$\mathbf{n} \to +\infty$$

Y normal

Multinomial distribution

a generalization of the binomial Distribution

$$f(x_1,\ldots,x_k;n,p_1,\ldots,p_k) = \Pr(X_1 = x_1 \text{ and } .. \text{ and } X_k = x_k)$$
$$= \begin{cases} \frac{n!}{x_1!\cdots x_k!} p_1^{x_1} \times \cdots \times p_k^{x_k}, & \text{when } \sum_{i=1}^k x_i = n \\ 0 & \text{otherwise} \end{cases}$$

$$f(x_1,\ldots,x_k;p_1,\ldots,p_k) = \frac{\Gamma(\sum_i x_i + 1)}{\prod_i \Gamma(x_i+1)} \prod_{i=1}^k p_i^{x_i} \quad \mathrm{E}(X_i) = np_i \quad \operatorname{var}(X_i) = np_i(1-p_i)$$

$$\operatorname{cov}(X_i, X_j) = -np_ip_j$$

Dirichlet distribution

a multivariate generalization of the beta distribution

$$f(x) = \begin{cases} \frac{1}{\beta(a,b)} x^{a-1}(1-x)^{b-1} & 0 \le x \le 1 \\ 0 & \text{otherwise} \end{cases}$$

$$f(x_1, \ldots, x_K; \alpha_1, \ldots, \alpha_K) = \frac{1}{\mathrm{B}(\boldsymbol{\alpha})} \prod_{i=1}^{K} x_i^{\alpha_i - 1}$$

$$X = (X_1, \ldots, X_K) \sim \mathrm{Dir}(\alpha)$$

$$\sum_{i=1}^{K} x_i = 1 \text{ and } x_i \ge 0 \text{ for all } i \in [1, K]$$

$$\mathrm{E}[X_i] = \frac{\alpha_i}{\alpha_0}$$

$$\mathrm{Var}[X_i] = \frac{\alpha_i(\alpha_0 - \alpha_i)}{\alpha_0^2(\alpha_0 + 1)}$$

$$\mathrm{B}(\boldsymbol{\alpha}) = \frac{\prod_{i=1}^{K} \Gamma(\alpha_i)}{\Gamma\left(\sum_{i=1}^{K} \alpha_i\right)}, \qquad \boldsymbol{\alpha} = (\alpha_1, \ldots, \alpha_K)$$

$$\mathrm{Cov}[X_i, X_j] = \frac{-\alpha_i \alpha_j}{\alpha_0^2(\alpha_0 + 1)}$$

Information Theory

Information Theory

- representing data in a compact fashion
- transmitting and storing it in a way that is robust to errors
- allocating short codewords to highly probable
  longer codewords to less probable bit strings
- Data mining

Information Theory

Information and Probability

- Information relates to possible outcomes of an event
- information is related the probability of an outcome
- The smaller the probability of an outcome, the more
- information it provides
- Entropy is the commonly used measure

Information Theory

- a variable (event), X,
- with n possible values (outcomes), x_1, x_2 …, x_n
- each outcome having probability, p_1, p_2 …, p_n
- the entropy of X , H(X), is given by

$$H(X) = -\sum_{i=1}^{n} p_i \log_2 p_i$$

- Entropy is between 0 and log2n and is measured in bits entropy is a measure of how many bits it takes to represent an observation of X on average

Information Theory

- Example:

  - For a coin with probability p of heads and probability q = 1 – p of tails

$$H = -p \log_2 p - q \log_2 q$$

For p= 0.5, q = 0.5 (fair coin) H = 1
For p = 1 or q = 1, H = 0

Information Theory

- a number of observations (m) of some attribute, X, e.g., the hair color of students in the class, where there are n different possible values
  the number of observation in the i^{th} category is m_i

$$H(X) = -\sum_{i=1}^{n} \frac{m_i}{m} \log_2 \frac{m_i}{m}$$

Hair Color	Count	p	$-p\log_2 p$
Black	75	0.75	0.3113
Brown	15	0.15	0.4105
Blond	5	0.05	0.2161
Red	0	0.00	0
Other	5	0.05	0.2161
Total	100	1.0	1.1540

Maximum entropy is log25 = 2.3219

Drawback of Correlation : Non-linear Relationships

- correlation is 0, no linear relationship between the attributes of the two data objects
- non-linear relationships may still exist.

Example:

$x = (-3, -2, -1, 0, 1, 2, 3)$

$y = (9, 4, 1, 0, 1, 4, 9)$

$y_i = x_i^2$

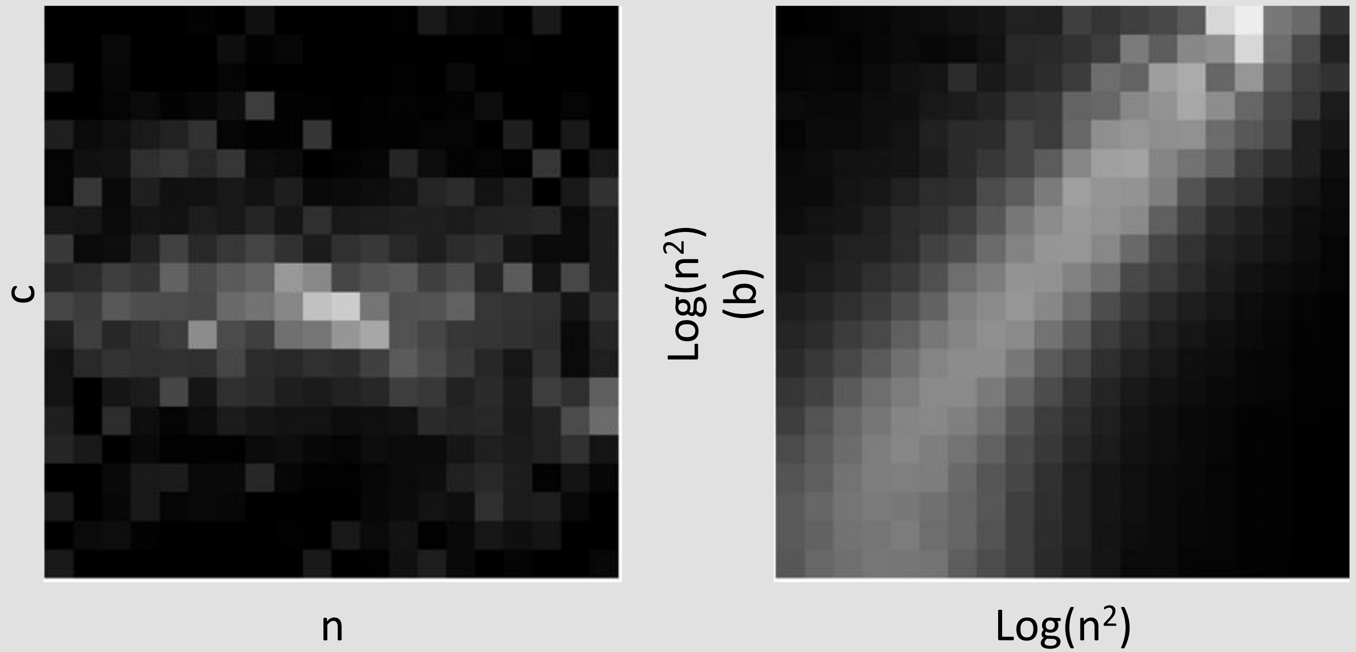
$\text{mean}(x) = 0$, $\text{mean}(y) = 4$

$\text{std}(x) = 2.16$, $\text{std}(y) = 3.74$

$\text{corr} = (-3)(5)+(-2)(0)+(-1)(-3)+(0)(-4)+(1)(-3)+(2)(0)+3(5) \,/\, (6 * 2.16 * 3.74)$

$= 0$

Drawback of Correlation : Non-linear Relationships



(a) (b)

Mutual Information

Information one variable provides about another

Formally, $\mathrm{I(X, Y) = H(X) + H(Y) - H(X, Y)}$, where

H(X,Y) is the joint entropy of X and Y,

$$H(X,Y) = -\sum_i \sum_j p_{ij} \log_2 p_{ij}$$

Where pij is the probability that the ith value of X and the jth value of Y occur together

how similar the joint distribution p(X, Y) is to the factored distribution

p(X)p(Y).

MI is zero iff the variables are independent

MI between X and Y as the reduction in uncertainty about X after observing Y

Mutual Information example

Student Status	Count	p	$-p\log_2 p$
Undergrad	45	0.45	0.5184
Grad	55	0.55	0.4744
Total	100	1.00	0.9928

Grade	Count	p	$-p\log_2 p$
A	35	0.35	0.5301
B	50	0.50	0.5000
C	15	0.15	0.4105
Total	100	1.00	1.4406

Student Status	Grade	Count	p	$-p\log_2 p$
Undergrad	A	5	0.05	0.2161
Undergrad	B	30	0.30	0.5211
Undergrad	C	10	0.10	0.3322
Grad	A	30	0.30	0.5211
Grad	B	20	0.20	0.4644
Grad	C	5	0.05	0.2161
Total		100	1.00	2.2710

Mutual information of Student Status and Grade = 0.9928 + 1.4406 - 2.2710 = 0.1624

Information Theory

KL divergence

One way to measure the dissimilarity of two probability distributions, p and q

$$\mathbb{KL}(p\|q) \triangleq \sum_{k=1}^{K} p_k \log \frac{p_k}{q_k}$$

Cross Entropy

$$\mathbb{KL}(p\|q) = \sum_k p_k \log p_k - \sum_k p_k \log q_k = -\mathbb{H}(p) + \mathbb{H}(p, q)$$

$$\mathbb{H}(p, q) \triangleq -\sum_k p_k \log q_k$$

cross entropy is the average number of bits needed to encode data coming from a source with distribution p when we use model q to define our codebook

Entropy : expected number of bits if we use the true model

KL divergence is the average number of extra bits needed to encode the data

$$\mathbb{KL}(p \| q) \geq 0 \text{ with equality iff } p = q$$

Information Theory

Mutual information

$$\mathbb{I}(X;Y) \triangleq \mathbb{KL}(p(X,Y) \| p(X)p(Y)) = \sum_x \sum_y p(x,y) \log \frac{p(x,y)}{p(x)p(y)}$$

$$\mathbb{I}(X;Y) \geq 0$$